

MUSTAFA SHOUKAT

Senior AI Engineer, Enterprise AI Architecture and Technical Leadership

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Executive Positioning Brief

Staff AI Engineer, Teradata, 24 September 2026

This brief maps my experience to the core requirements of the Staff AI Engineer role. Each area gives the requirement, what I delivered and the result.

AI PLATFORM ARCHITECTURE

- **Requirement.** Design and deliver highly scalable AI products built for reliability and performance.
- **Delivered.** Designed ByanRAG 2.2 end to end: FastAPI services, isolated vector collections per organization, a five level access model enforced on every endpoint, full audit logging and a 14 panel admin console, packaged for on premises and air gapped deployment.
- **Result.** Two organizations and more than 150 users share one platform with strict data isolation inside regulated networks.

RETRIEVAL AUGMENTED GENERATION

- **Requirement.** Apply LLMs and retrieval to enterprise data with answers people can trust.
- **Delivered.** Built hybrid retrieval that combines dense and BM25 search with Reciprocal Rank Fusion and CrossEncoder reranking, tuned for bilingual Arabic and English content and measured with Recall@5, MRR and faithfulness scoring.
- **Result.** Source cited answers in 2 to 3 seconds on self hosted open source models.

ORCHESTRATION AND AGENTIC AI

- **Requirement.** Deliver agentic AI and large language model systems.
- **Delivered.** Led multi agent workflows in LangGraph and CrewAI with tool calling and multi step planning: Global Holman agents read SAM.gov solicitations, analyze requirements and draft proposals, and the Whoza.ai voice agent answers and qualifies calls. Built my own MCP server with Arabic tools and use MCP tools in daily development.
- **Result.** Analyst work that took days is now automated on a provider agnostic LLM layer.

DISTRIBUTED SYSTEMS

- **Requirement.** Apply distributed system patterns with strong data consistency.
- **Delivered.** Use Celery and Redis for background ingestion and agent jobs, async PostgreSQL, streaming over WebSockets and server sent events, and idempotent, signature verified webhooks. Moved VacantSeek 2.0 onto AWS ECS with CodeBuild pipelines and S3 storage.
- **Result.** Work continues when a worker fails, and retries never create duplicate records.

PRODUCTION READINESS

- **Requirement.** Resolve performance bottlenecks and production issues and maintain code quality.
- **Delivered.** Run services in Docker behind nginx with TLS and CI/CD, with more than 600 tests on ByanRAG, monitoring in Prometheus, Grafana and Loki, tracing in LangSmith and Sentry, and guardrails against prompt injection. Led more than 340 merged pull requests of hardening on VacantSeek 2.0.
- **Result.** Issues surface on dashboards before users report them; the same evaluation discipline helped take Yehia R1 to first place on AraGen.

TECHNICAL LEADERSHIP

- **Requirement.** Mentor engineers and influence architecture decisions.
- **Delivered.** Lead and mentor the BryanRAG team of four, own its architecture decisions and set its standards for testing, deployment and code review. Trained more than 150 professionals and work directly with clients in four countries.
- **Result.** The team ships to one shared standard.

ENGINEERING FOUNDATIONS

Python (primary), Java, TypeScript and SQL; RESTful APIs in FastAPI; PostgreSQL, MongoDB, Redis and pgvector; AWS, GCP and Azure; Docker and AWS ECS; GitHub Copilot and MCP based coding assistants. My orchestration and messaging patterns from Docker, ECS, Celery and Redis carry directly to Kubernetes and Kafka.